IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 31 October 2024, accepted 18 November 2024, date of publication 29 November 2024,
date of current version 13 December 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3508796

RESEARCH ARTICLE

Image Segmentation Method With Improved GA Optimization of Two-Dimensional Maximum Entropy

JIN WANG[1,2], YANLI TAN[2], XIAONING BO[1], AND GUOQIN LI[2]
[1]Department of Electronic Engineering, Taiyuan Institute of Technology, Taiyuan 030008, China
[2]College of Computing, Informatics and Mathematics, Universiti Teknologi MARA (UiTM), Selangor 40450, Malaysia

Corresponding author: Jin Wang (wangj@tit.edu.cn)

This work was supported by the Scientific and Technological Innovation Programs of Higher Education Institutions in Shanxi under Grant STIP 2022L533, Grant 2022L534, and Grant 2022L538.

ABSTRACT Image segmentation is the cornerstone that determines the effectiveness of image processing, but traditional image segmentation methods have issues such as long computation time, low recognition accuracy, and poor anti-interference ability. To address this issue, research improves the genetic algorithm using adaptive hybridization and adaptive mutation probability, and combines it with the Bat algorithm to optimize the local optimization problem of the image. The sparrow algorithm is utilized to optimize the two-dimensional maximum entropy of the image, and the nonlinear inertia weight factor is brought to optimize the local search ability. The Levy flight constant is used to overcome the local optimization problem. The experiment findings indicate that the optimized algorithm improves the similarity of medical image features by an average of 11.2%, reduces segmentation accuracy by 2.6% under noise interference compared to other algorithms, and has an average peak signal-to-noise ratio 0.96 higher than other algorithms. From this, the improved algorithm greatly raises the similarity of segmented image features, has stronger resistance to noise interference than other algorithms, and significantly improves the recognition accuracy of different parts of the image. The improved algorithm provides a reference for subsequent image processing research.

INDEX TERMS Genetic algorithm, adaptive mutation probability, sparrow algorithm, image segmentation, two-dimensional maximum entropy.

I. INTRODUCTION

Image processing technology originated in the 1920s, when a photograph was transmitted from the UK to the US using digital compression technology via submarine cables [1], [2]. With the development of the times, the application of information technology has become increasingly widespread. Image processing technology has become an important means for humans to obtain, transmit, and express information, but the requirements for image processing technology are also increasing [3], [4]. Image processing technology has wide applications in remote sensing satellite images, radar images, communication processes, medical images, and military fields [5]. Image processing techniques include image transformation, encoding and compression, enhancement and restoration, segmentation, description, and classification and recognition. Image segmentation (IS) is the most pivotal process [6]. Regarding the issue of IS accuracy, Cheng G et al. proposed a holistic prototype activation network to solve the problems such as over-fittingthat existing methods are prone to when segmenting fewer shots. This network filtered un-managed categories by high confidence, and used a cross-reference decoder for interactive feature re-weighting and multi-level feature aggregation. Experiments showed that the network had high accuracy and precision in performing weakly labeled and zero-shot segmentation, as well as high flexibility [7]. Lang C et al. addressed the bias problem of less-shot segmentation under low data conditions by proposing a new basis and meta. The method applied a base learner to a meta learner to recognize base class objects

The associate editor coordinating the review of this manuscript and approving it for publication was Tai Fei.

© 2024 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

and adaptively integrated the parallel outputs. Experiments showed that the method significantly outperformed other methods on standard basis tests [8]. Lang C et al. proposed a new flexible and generalized framework to improve the success rate of minority lens segmentation. The framework used a self-inference scheme to divide the coarse segmentation mask into multi-regions with different attributes, and utilized the mask averaging pooling technique to quickly generate images that support induction. Experiments showed that the framework had a high segmentation success rate with complete segmented objects and clear boundaries [9].

To raise the accuracy of IS and lessen computational costs, Abdel-Basset M et al. raised a new IS method with the ocean predator algorithm. This method avoided local optima by gradually moving poor solutions to the optimal solution (OS), and randomly searched in space with a certain probability. Experiments denoted that this method had higher IS accuracy and computational cost than other methods when the image threshold level was high [10]. Gharehchopogh FS et al. developed an optimized African vulture algorithm to solve the objective problems of different mechanisms and improve algorithm performance. This algorithm used three binary thresholds in multi-threshold IS and increased the population diversity by a quantum rotation gate mechanism. Experiments showed that this algorithm could effectively avoid falling into local optimal constraints and had higher IS accuracy [11]. Khan AA et al. developed an image recognition model with genetic algorithm (GA) to raise the accuracy and computational speed of facial image recognition. This model used principal component analysis to reduce the dataset dimensionality and optimized the search pattern for image matching using GAs and neural networks. The experiment indicated that the facial image recognition accuracy of the model was 93.7%, and the model had high stability [12]. Wu D et al. proposed an image threshold segmentation (ITS) method with an optimized sparrow search algorithm and two-dimensional maximum entropy (2DME) to improve the ITS performance. This method improved local search capability by introducing nonlinear inertia weights (NIWs) and using Levy flight to solve the local OS problem in the later stages of iteration. Experiments showed that this method improved segmentation accuracy on various medical images in terms of peak signal-to-noise ratio (PSNR) and feature similarity [13]. Alhijawi B et al. developed a new differential evolution adaptive Harris Hawkes algorithm to address the limited exploration capability of the Harris Hawk optimization method. This algorithm limited the escape energy during the exploration phase to a certain range, used differential evolution to improve exploration capability, and used a 2DME function to obtain contextual information. The experiment showed that the IS ability of this algorithm was greatly better than other algorithms [14].

In summary, existing research methods have explored issues such as IS accuracy and computational speed from multiple perspectives, and have achieved certain results. However, IS accuracy can no longer meet the growing demand. Therefore, the study improves the GA to optimize the 2DME, innovatively introduces adaptive hybridization and mutation probabilities, reduces the probability of local optima, and uses NIW factors and Levy flight constants to strengthen the algorithm's local search ability and improve IS accuracy. The improved algorithm aims to improve the accuracy of image feature recognition, enhance anti-interference ability, and improve computing speed.

II. METHODS AND MATERIALS

A. IMAGE SEGMENTATION BASED ON IMPROVED GA

Traditional GAs suffer from high time costs and the tendency to fall in local optima. Existing GA improvement methods typically use waterfall line method and volume exclusion chromatography to change the selection and hybridization methods of genetic individuals, exchanging information fragments from different individuals randomly [15]. However, in medical images, there are usually characteristics such as low contrast, low SNR, low lighting intensity, and significant differences between individuals. During information exchange, some individual information is lost, resulting in insufficient segmentation accuracy of image information and even erroneous segmentation [16]. The main operating steps of GAs include individual evaluation, selection operation, hybridization operation, and mutation operation. Existing GAs usually have only one selection method for selection operation. In this method, the probability of selecting individuals is proportional to their FV, which can easily lead to some individuals being repeatedly selected and those with high FVs monopolizing the next generation [17]. The study uses a combination of random traversal selection and tournament selection for individual selection operations and divides the selection wheel based on the FV of each individual. After rotating the wheel, multiple selection points are provided to select multiple individuals at once. It randomly chooses n individuals from the remaining population, chooses the best individual to enter the next generation, and puts the remaining individuals back into the population. The combined method not only ensures the continuation of excellent individuals in the population, but also provides opportunities for individuals with low fitness to be selected, ensuring the diversity of the population. The study uses a combination of GA and Bat algorithm to improve the algorithm's optimization searching accuracy by using adaptive selection, crossover, and mutation operators to select, crossover, and mutate Bat individuals after they have performed a position update before proceeding to the next iteration.

To further raise the population's diversity and find the optimal path, it is necessary to perform hybridization and mutation operations on the population, and the hybridization probability is related to the degree of population diversity. The greater the population diversity, the higher the probability of finding the optimal path, but the probability of local OSs will also increase accordingly. If the hybridization probability

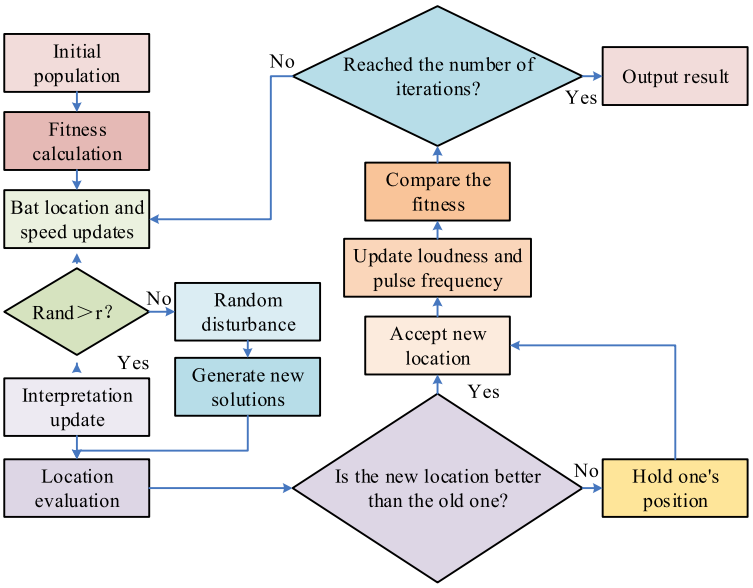


FIGURE 1. Improved GA combined with Bat algorithm.

is too low, it will affect the search efficiency of the population. Thus, it will determine the size of the hybridization probability based on the actual situation. In the early stage of algorithm iteration, first is to increase the search range of the population, select a higher hybridization probability, and quickly improve population diversity. In the later stage of iteration, selecting a smaller hybridization probability ensures population stability, increases local search ability, and improves the algorithm's convergence speed. The adaptive hybridization probability calculation is denoted in equation (1) [18].

$$p_{hs} = \frac{2f_{\max} - \bar{f} - f(n_1, n_2)}{f_{\max} - f_{\min}} \tag{1}$$

In equation (1), p_{hs} represents the adaptive hybridization probability considering only fitness. $f_{\max}$ means the max FV in the population. $\bar{f}$ means the average FV in the population. $f(n_1, n_2)$ represents the maximum FV between the two individuals to be hybridized. $f_{\min}$ represents the minimum FV in the population. According to different iteration periods, the adaptive hybridization probability is optimized and calculated as shown in equation (2) [19].

$$p_h = \frac{2f_{\max} - \bar{f} - f(n_1, n_2)}{f_{\max} - f_{\min}} \cdot \frac{I - i}{I} \tag{2}$$

In equation (2), p_h means adapts the hybridization probability. I means the total amount of iterations. i means the current amount of iterations. Similarly, in the early iteration, a larger mutation probability is needed to increase the breadth of the population and raise the global search ability. In the later stages of iteration, the mutation probability is reduced to improve convergence speed. The adaptive mutation probability calculation is shown in equation (3) [20].

$$p_v = \frac{2f_{\max} - \bar{f} - f}{20(f_{\max} - f_{\min})} \cdot \frac{I - i}{I} \tag{3}$$

In equation (3), p_v means the adaptive mutation probability, and f represents the fitness of the current individual. GAs have problems such as difficult direction control and susceptibility to getting stuck in local optima, while Bat algorithms initialize a set of random solutions, iterate to explore the OS, and randomly fly around the OS to generate local new solutions, enhancing local search capabilities. GA and Bat algorithm are integrated to improve and optimize the shortcomings of GA. The improved GA running process is indicated in Figure 1.

In Figure 1, the initial population of the GA is the same as that of the Bat algorithm. The FV of the population is calculated and the OS is saved. The position and velocity information of all Bat individuals are updated. It should determine the relationship between the generated random number and the frequency of sound waves of individual Bats at the current time. If the random number is larger, it will evaluate the new location. If the frequency of sound waves is larger, it will randomly perturb near the OS to generate a new solution and then evaluate the new location. It should determine whether the random number generated by the new location is greater than the sound volume and whether the new location is better than the original location. If both conditions are met, it will store the new location information and update the sound frequency and sound volume. If they cannot be met simultaneously, it will maintain the original location. Individual selection, hybridization, and mutation operations are performed on the updated Bat population using GA to generate new offspring Bat individuals. The fitness levels of the parent and offspring individuals are compared, the

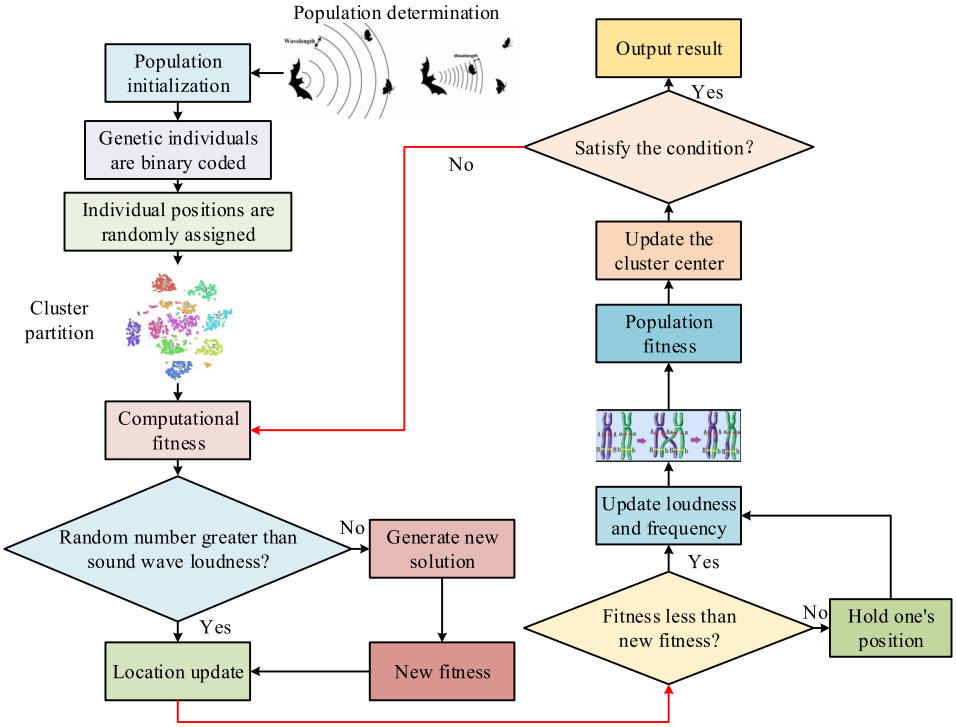


FIGURE 2. Image clustering segmentation based on improved GA.

optimal position is selected, and whether the iteration count or fitness requirements have been reached is determined. If yes, it will end the algorithm process and output the results. If not, it will repeat the above operations until the conditions are met. The optimized GA is used for image clustering and segmentation, as shown in Figure 2.

In Figure 2, the population size and objective function are first determined based on the actual situation, and the population is initialized. The algorithm ends with the condition that the max amount of iterations is reached and the average fitness ratio of the parent and child generations is between 1 and 1.005. It inputs the adaptive hybridization probability p_h and adaptive mutation probability p_v, while binary encoding the individuals of each initial population. It randomly allocates the positions of all individuals and uses them as clustering centers to calculate the fitness of all individuals, retaining the OS that appears in the current state. The position and velocity information of all individuals are updated, and a numerical value is generated. If the value is greater than the frequency of the sound wave, it will perturb the optimal position to produce a new solution, and continue to generate a new value. If the value is less than the loudness of the sound wave and the fitness of the parent generation is less than that of the child generation, it will update the position of the individual, and update the volume and frequency of the emitted sound wave of the individual. It will select, hybridize, and mutate the updated individuals, calculate the fitness of the offspring population, determine whether the population fitness meets the requirements for updating the clustering center, perform mean clustering on the image, and judge whether the population meets the above algorithm termination conditions. If it does, it will output the result.

B. OPTIMIZATION OF 2D MAXIMUM ENTROPY IMAGE THRESHOLD SEGMENTATION BASED ON SPARROW ALGORITHM

ITS is the process of determining the segmentation threshold by judging the difference in grayscale values between the image object and background, and then comparing the grayscale values of pixels in the image with the threshold to determine their ownership and complete the segmentation of the image. When selecting image thresholds, the sparrow algorithm is used. The SSA has the merits of simplicity, easy implementation, fast solving speed, high computational accuracy, and high robustness. However, it is easy to fall in local optima and has many limitations on parameter selection [21]. Nonlinear inertial weight factors are introduced to enhance the follower's search range and more search directions, optimize the local search capability, and Levy's flight constants are adopted for the vigilant's position update to get rid of the local optimum problem. In the sparrow algorithm, individuals are roughly divided into three types: searchers, followers, and alert individuals. Searchers occupy more computing resources in the algorithm and are mainly responsible for searching for food. Followers are responsible for traveling to the area after searchers discover food. After discovering danger, both searchers and followers will

temporarily transform into alert individuals to remind other individuals. The update of the searcher's location information is shown in equation (4) [22].

$$\begin{cases} S_{i,j}^{x+1} = S_{i,j} \cdot \exp\left(-\dfrac{i}{\alpha \cdot it_{\max}}\right), B < SV \\ S_{i,j} + Q \cdot L, B \geq SV \end{cases} \tag{4}$$

In equation (4), $S_{i,j}^{x+1}$ means the updated searcher position. i means the amount of individuals. j means the dimensional information of individuals. $S_{i,j}$ means the current position information of individuals. α means a random number between 0-1. $it_{\max}$ represents the max amount of iterations. B represents the alert value between 0-1. SV represents the safety value between 0.5-1. Q represents a random number. L represents a matrix with length 1 and width of dimensionality, and all numbers in the matrix are 1. Followers have low energy storage values and cannot complete the search task alone. Therefore, they go to the area where food exists after the searcher sends a signal, and the follower's location information is updated as shown in equation (5) [23].

$$\begin{cases} S_{i,j}^{x+1} = Q \cdot \exp\left(\dfrac{S_w - S_{i,j}}{i^2}\right), i > n/2 \\ S_b + \left|S_{i,j} - S_b\right| \cdot L' \cdot L, i \leq n/2 \end{cases} \tag{5}$$

In equation (5), S_w represents the individual farthest from the center in the population. n means the total amount of individuals in the population. S_b means the individual closest to the center in the population. L' represents a matrix with a length of 1 and a width of dimensionality. The number in the matrix is a random number of -1 or 1. When $i > n/2$ occurs, the followers will move toward the searcher to obtain food and improve their own ability reserve. However, the matrix L' that guides the follower's direction only has two values, -1 and 1, which limits the follower's ability to search for the OS in the local area and leads to the algorithm entering the local OS trap in the later stage of iteration. Therefore, when updating the position of the follower, an NIW factor is added to optimize the search direction of the follower, as shown in equation (6) [24].

$$\lambda = \lambda_{\max} - (\lambda_{\max} - \lambda_{\min}) \cos\left(\frac{X \cdot \pi}{it_{\max}}\right) \tag{6}$$

In equation (6), λ represents the NIW factor. $\lambda_{\max}$ means the max value of the NIW factor. $\lambda_{\min}$ means the mini value of the NIW factor. X means the current iteration count. After adding the NIW factor, when the follower searches for the OS near the searcher, the searchable area becomes larger and the search becomes more flexible, effectively enhancing the local search ability and improving the diversity of the population. When updating the position of the alert, the step size control parameter is a normally distributed random number with a symmetrical shape of high in the middle and low on both sides, which can interfere with the algorithm's ability to get rid of local optima. Therefore, Levy flight random constant

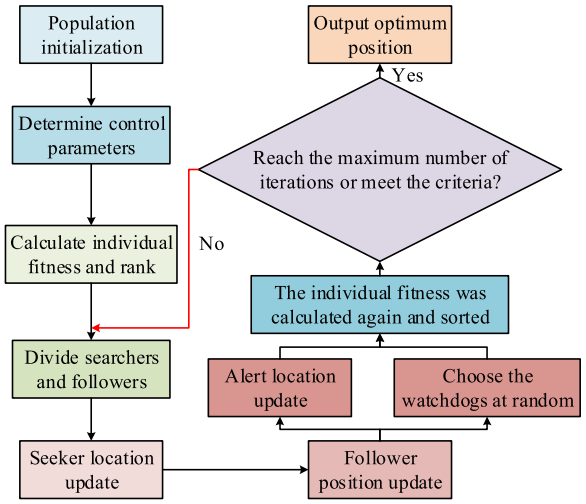


FIGURE 3. Operation flow of sparrow algorithm after optimization.

is introduced and calculated as shown in equation (7) [25].

$$\begin{cases} Levy = \dfrac{\delta_k \cdot k}{|g|^{\frac{1}{\gamma}}} \\ \delta_k = \left[\dfrac{\Gamma(1+\gamma)\sin\frac{\gamma\pi}{2}}{\Gamma\left(\frac{1+\gamma}{2}\gamma \cdot 2^{\frac{\gamma-1}{2}}\right)}\right] \\ \delta_g = 1 \end{cases} \tag{7}$$

In equation (7), k and g respectively mean the random numbers that follows the normal distribution of $k \sim N\left(0, \delta_k^2\right)$ and $g \sim N\left(0, \delta_g^2\right)$. γ represents a step size random number. The optimized alert position update formula is shown in equation (8) [26].

$$\begin{cases} S_{i,j}^{x+1} = S_b^x + Levy \cdot \left|S_{i,j}^x - S_b^x\right|, f_i > f_g \\ S_{i,j}^{x+1} = S_{i,j}^x + K\left(\dfrac{S_{i,j}^x - S_w^x}{(f_i - f_w) + \varepsilon}\right), f_i = f_g \end{cases} \tag{8}$$

In equation (8), f_i means the fitness function value of the i-th individual. f_g and f_w mean the fitness function values of the best and the worst individuals in the current offspring, respectively. K means a random number between -1 and 1. ε means a smaller constant. According to equation (8), if the fitness function value of the current alert individual is greater than the fitness function value of the optimal individual, the individual's position is located at the edge of the population and will move towards the position of the optimal individual [27]. After using the Levy random constant, the alert will once escape from the optimal region after multiple movements, thus solving the problem of local optima caused by searching only locally. The optimized sparrow algorithm running process is shown in Figure 3.

In Figure 3, first is to determine the relevant parameters of the algorithm, initialize the population, and then calculate the fitness function values of all individuals and sort them from small to large according to their FVs. According to the

individual's FV, the ratio of searchers to followers is divided in a ratio of 7:3, and the position of searchers is updated. At the same time, a formula that incorporates NIW factors is used to update the position of followers. It randomly selects a portion of alert individuals based on their 20% proportion, or updates their positions using a formula that incorporates the Levy random constant. It recalculates the fitness of all individuals at the new location and sorts them by size to determine if the max amount of iterations has been reached or if the stopping condition is met. If the condition is satisfied, it will output the best position information. If not, it will repeat the above operation until the condition is met.

The one-dimensional max entropy threshold segmentation method is commonly used in IS. Although the segmentation speed is fast, it only considers the grayscale values and distribution of each pixel in the plane, and cannot reflect the spatial information distribution of the image. The 2DME segmentation method can simultaneously reflect the information of pixels in the image and the grayscale distribution information of other points around the pixels [28]. The 2DME ITS method divides the image into four regions through two sets of thresholds, namely the grayscale threshold p and the neighborhood threshold q. The diagonal distribution of the first and third quadrant regions represent the target region and background region, respectively. The second and fourth quadrant regions represent the interference information and edge information of the image. The 2DME ITS method aims to find the optimal segmentation threshold based on the grayscale information of the pixels in the first and third quadrant regions. The area with a side length of 3 around each pixel in the image is taken as the neighborhood of that pixel, and the average grayscale value of each pixel's neighborhood is calculated as shown in equation (9) [29].

$$C_{ij} = \frac{n_{ij}}{xy} \tag{9}$$

In equation (9), C_{ij} represents the probability that the average gray value of a pixel's neighborhood is equal to j when the gray value is equal to i. n_{ij} represents the total amount of pixels in the pixel's neighborhood when the gray value is equal to i and the average gray value of the pixel's neighborhood is equal to j. xy means the pixel size of the image. The probability of the optimal threshold in the target area and background area is indicated in equation (10) [30].

$$\begin{cases} C_g(p,q) = \sum_{i=0}^{p-1}\sum_{j=0}^{q-1} C_{ij} \\ C_b(p,q) = \sum_{i=p}^{L-1}\sum_{j=q}^{L-1} C_{ij} \end{cases} \tag{10}$$

In equation (10), $C_g(p,q)$ means the probability of the optimal segmentation threshold being in the target area. $C_b(p,q)$ means the probability of the optimal segmentation threshold being in the background area, and $L = 256$ represents the image grayscale limit. Because the image contains less interference and edge information, it can be ignored during calculation. The entropy calculation of the target area is shown in equation (11).

$$H_g(p,q) = -\sum_{i=0}^{p-1}\sum_{j=0}^{q-1} \frac{C_{ij}}{C_g(p,q)} \ln \frac{C_{ij}}{C_g(p,q)} \tag{11}$$

In equation (11), $H_g(p,q)$ represents the entropy of the target area, and the entropy calculation of the background area is shown in equation (12).

$$H_b(p,q) = -\sum_{i=p}^{L-1}\sum_{j=q}^{L-1} \frac{C_{ij}}{C_b(p,q)} \ln \frac{C_{ij}}{C_b(p,q)} \tag{12}$$

In equation (12), $H_b(p,q)$ represents the entropy of the background region, and it can be concluded that the calculation of the maximum entropy in two dimensions is shown in equation (13).

$$H(p,q) = H_g(p,q) + H_b(p,q) \tag{13}$$

In equation (13), $H(p,q)$ means the 2DME of the image. In the image, the larger the value of the 2DME, the more information it contains, and the optimal threshold required should also be larger. Therefore, the relationship between the optimal threshold and the 2DME is shown in equation (14).

$$(p,q)_{best} = \arg\max\left(0 \le p \le L-1, 0 \le q \le L-1\right)[H(p,q)] \tag{14}$$

In equation (14), $(p,q)_{best}$ represents the optimal threshold of the image, and 2DME is used as the fitness function of the sparrow algorithm to search for the OS. The calculation of searching for the optimal threshold is shown in equation (15).

$$find\ (p,q)_{best} = \arg\max\left(0 \le p \le L-1, 0 \le q \le L-1\right)[H(p,q)] \tag{15}$$

The ITS process using the 2D Maximum Entropy-Sparrow Search Algorithm (2DME-SSA) is shown in Figure 4.

In Figure 4, relevant information is extracted from the image to be recognized, and the grayscale histogram of the image is read. The population of the sparrow algorithm is initialized to determine the relevant control parameters. It calculates the FVs of all individuals in the population and uses the sparrow algorithm for optimization to find the optimal ITS. It should determine whether the latest optimal segmentation threshold meets the stopping condition. If it meets the condition, it will output the optimal threshold and segment the image. If it does not meet the condition, it will calculate the FV of the individual again and repeat the above operation until the condition is met or the amount of iterations is reached. To identify the feasibility of the improved algorithm in IS, five different test functions are selected. The expressions, value ranges, dimensions, and optimal FVs of the test functions are denoted in Table 1.

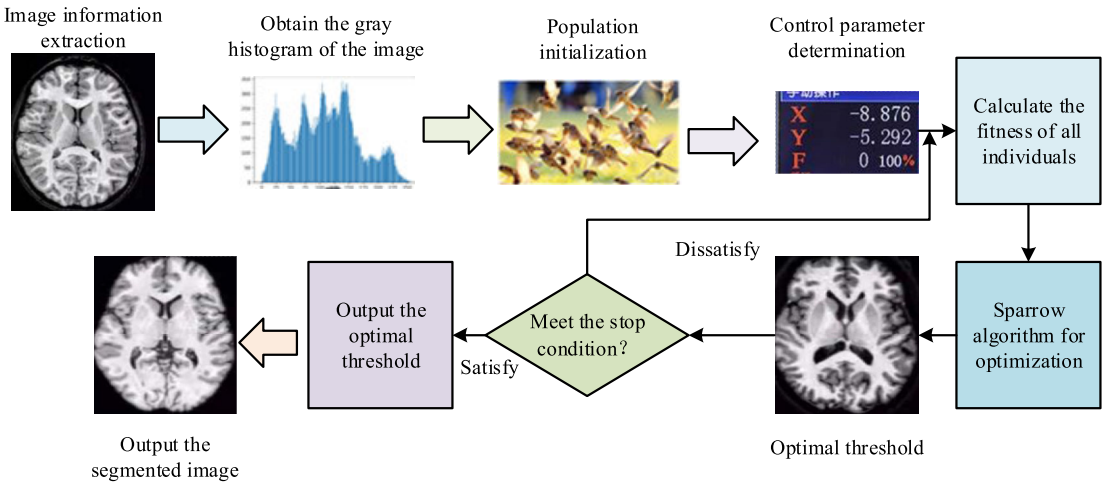


FIGURE 4. Segmentation process of 2DME-SSA.

TABLE 1. Test function for feasibility calculation of the improved Algorithm.

Function expression	Value ranges	Dimensions	Optimal FVs
$F_1(x)=\sum_{i=1}^{n}x_i^2$	-80~80	30	0
$F_2(x)=\sum_{i=1}^{n}\left(\left\lvert x_i+\frac{1}{2}\right\rvert\right)^2$	-80~80	30	0
$F_3(x)=-\sum_{i=1}^{4}c_i\exp\left(-\sum_{j=1}^{3}a_{ij}\left(x_j-p_{ij}\right)^2\right)$	1~5	3	0.024
$F_4(x)=\sum_{i=1}^{n}\left(\sum_{j=1}^{i}x_j\right)^2$	-30~30	30	0
$F_5(x)=\sum_{i=1}^{n}\left[x_i^2-10\cos(2\pi x_i)+10\right]$	-5~5	30	0

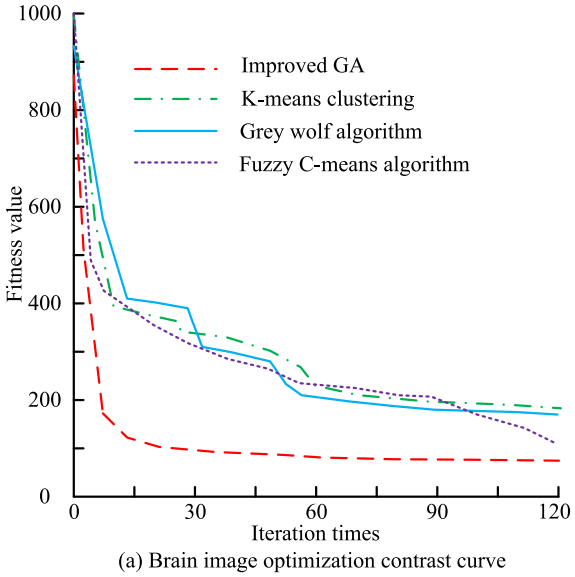
III. RESULTS

A. EXPERIMENTAL ANALYSIS OF IMAGE SEGMENTATION BASED ON IMPROVED GA

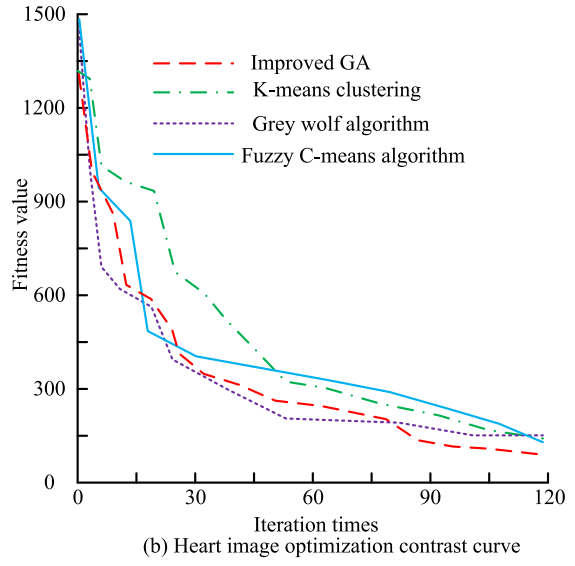
The algorithm running environment was Windows 10, the computer processor was Intel Core i5-8400 with a benchmark frequency of 2.80 GHz, and the test software was MATLAB R2018a. The IS experiment of improving GA adopted Brain2021 brain tumor medical imaging dataset and ACDC cardiac imaging dataset, which contained 750 and 150 images respectively. The max iteration times of the algorithm was set to 120, the frequency was set between 0.1 and 2, the sound volume was set to 0.2, the sound frequency was set to 1, and the individual velocity was set between -5 and 5. The comparative algorithms were K-means, K-means based on grey wolf algorithm, fuzzy C-means algorithm, and K-means with improved GA. The comparison of optimization curves between the two images is shown in Figure 5.

In Figure 5 (a), the optimization curve of the improved GA decreased the fastest, completing most of the optimization tasks in 0-20 iterations. The FV decreased from 870 to 41, a decrease of 829. The other three algorithms reached their optimal FVs after 60, 56, and 116 iterations, respectively. The optimal FVs of the improved GA were 145, 128, and 53 lower than the other three algorithms, respectively. In Figure 5 (b), the descent speed of the optimization curve of the improved GA slowed down, and it only reached the optimal FV after 88 iterations, with the FV decreasing from 1294 to 136, a decrease of 1158. The other three algorithms reached their optimal FVs at 101, 52, and 117 iterations, respectively. The optimal FVs of the improved GA were 22, 35, and 19 lower than the other three algorithms, respectively. The comparison of feature similarity among different parts after clustering segmentation is shown in Figure 6.

In Figure 6 (a), the improved GA had the highest feature similarity in the segmented image, reaching 64% feature

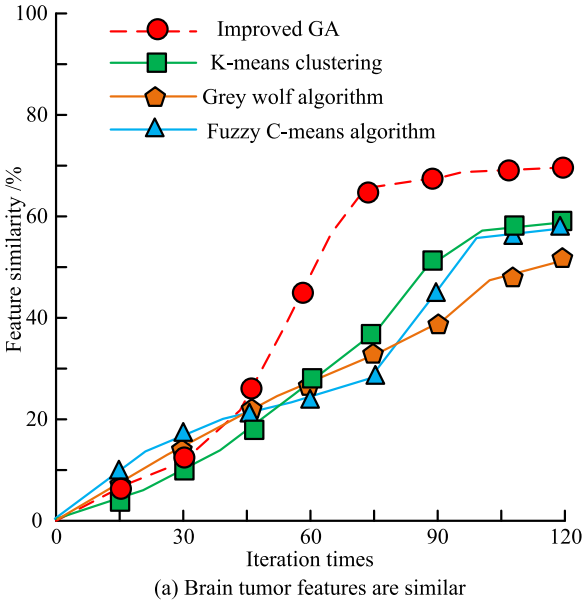


(a) Brain image optimization contrast curve

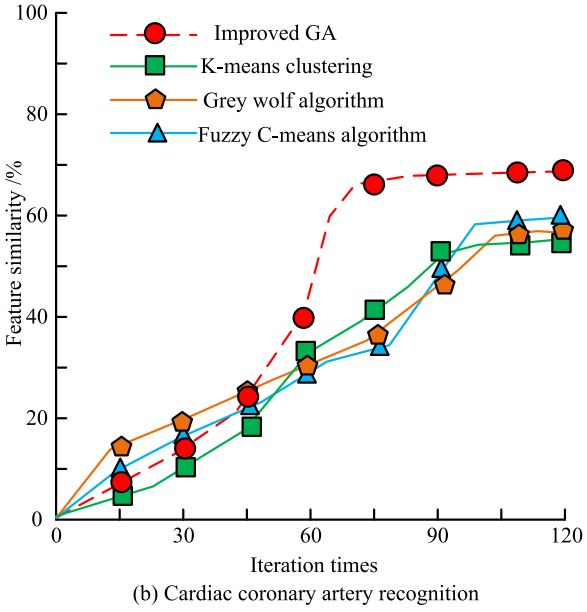


(b) Heart image optimization contrast curve

FIGURE 5. Comparison of medical image optimization curves of different parts.



(a) Brain tumor features are similar



(b) Cardiac coronary artery recognition

FIGURE 6. Feature similarity of different parts.

similarity after 74 iterations. The feature similarity increased the fastest between 46 and 74 iterations, and the convergence area of other algorithms was basically around 102 iterations. The maximum similarity of brain tumor features was 69%, which was 11%, 17%, and 12% higher than the similarity of brain tumor features of the other three algorithms, respectively. In Figure 6 (b), the similarity of image features after improved GA segmentation basically converged after 71 iterations, while the convergence area of other algorithms was around 98 iterations. The maximum similarity of coronary artery features in the heart was 67%, which was 13%, 10%, and 8% higher than the similarity of coronary artery features in the other three algorithms, respectively. The average FVs and average segmentation accuracy of different algorithms are denoted in Table 2.

In Table 2, the average FVs of the brain images using the improved GA were 52, 37, and 35 lower than the other three algorithms, respectively. The average segmentation accuracy of the improved algorithm was the highest at 83.4%. After adding Gaussian noise and exponential noise to the images, the segmentation accuracy decreased by 1.9% and 2.5%, respectively. After adding noise interference, the IS accuracy of the improved GA decreased less than other algorithms, indicating that the improved GA had stronger anti-interference ability. The average FVs of heart images using the improved GA were 43, 9, and 32 lower than the other three algorithms, respectively. The improved GA had the highest average segmentation accuracy in all three cases.

TABLE 2. Average fitness values and average segmentation accuracy of different Algorithms.

	Algorithm	Improved GA	K-means	Grey wolf algorithm	Fuzzy C-means algorithm
Brain image	Mean fitness	184	236	221	219
	Average segmentation accuracy/%	83.4	79.6	81.5	80.2
	Gaussian noise /%	81.5	75.1	78.2	76.5
	Exponential noise /%	80.9	72.6	77.9	75.4
Cardiac image	Mean fitness	305	348	314	337
	Average segmentation accuracy/%	74.6	69.5	72.7	71.6
	Gaussian noise /%	73.1	65.2	72.2	68.9
	Exponential noise /%	71.9	64.7	69.4	67.3

However, when Gaussian noise interference was added, the IS accuracy based on K-means using the grey wolf algorithm decreased less than that of the GA, indicating that the grey wolf algorithm had strong anti-interference ability in heart IS.

B. EXPERIMENTAL ANALYSIS OF OPTIMIZED 2DME-SSA

The experiment used PASCAL-PART ordinary image dataset and LNDb pulmonary image dataset for IS. The 2DME-SSA algorithm was compared and analyzed with PSO, Grey Wolf Optimizer (GWO), and Jellyfish Search (JS) algorithm for ITS. The test results of different algorithms on the benchmark function are indicated in Table 3.

In Table 3, the accuracy of the mean and variance obtained by the 2DME-SSA was the highest in each test function, with the lowest mean accuracy reaching E-9 and the lowest variance accuracy reaching E-5. Among all test functions, the optimal value of 2DME-SSA could obtain the global OS of 0 before the iteration was completed. Among other algorithms, only GWO algorithm could obtain the global OS of 0 in the first three benchmark functions. But the average accuracy and variance accuracy of all other algorithms were significantly lower than those of the 2DME-SSA. The convergence curves of different algorithms were drew using two benchmark functions, as shown in Figure (7).

In Figure 7, the vertical axis expresses the FV after logarithmic operation. In Figure 7 (a), 2DME-SSA achieved the maximum convergence accuracy after 40 iterations, and the convergence speed was significantly faster than other algorithms. In Figure 7 (b), 2DME-SSA quickly solved the convergence value after 95 iterations, and the final convergence values were 274, 325, and 338 lower than other algorithms, respectively. The comparison of the complexity of different algorithms for the benchmark function is shown in Figure 8.

In Figure 8 (a), the 2DME-SSA had the highest testing accuracy among all algorithms, achieving maximum testing accuracy after 15 iterations. Other algorithms required 52, 67, and 114 iterations, respectively, but the algorithm took 0.08 seconds slower to run than the fastest algorithm. In Figure 8 (b), the 2DME-SSA achieved a maximum testing accuracy of 78% after 27 iterations, while the others required 65, 82, and 86 iterations, respectively. The running time was 0.07 seconds slower than the fastest algorithm. The comparison of the optimal threshold and FVs of ordinary images measured using different algorithms is shown in Figure 9.

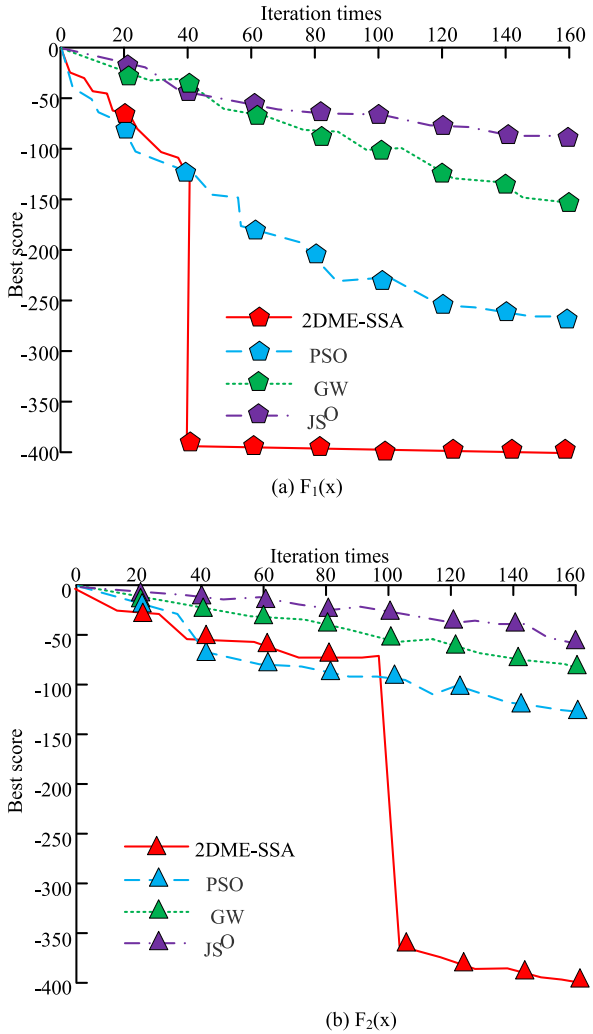


FIGURE 7. Convergence curves comparison of different algorithms.

TABLE 3. Test results of different benchmark functions.

Algorithm	Reference function	Mean value	Optimal value	Variance
2DME-SSA	$F_1(x)$	1.253E-71	0	4.528E-108
PSO		37.245	6.427	209.687
GWO		1.582E-64	0	2.057E-64
JS		1.735E-16	0	2.689E-18
2DME-SSA	$F_2(x)$	5.429E-69	0	7.529E-94
PSO		2.907	0.825	0.735
GWO		2.052E-54	0	1.958E-65
JS		1.843E-32	1.256E-15	4.287E-56
2DME-SSA	$F_3(x)$	8.254E-98	0	6.258E-47
PSO		376.582	185.362	1580.364
GWO		6.245E-77	0	7.563E-34
JS		3.058E-48	0.983	126.358
2DME-SSA	$F_4(x)$	1.358E-9	0	2.561E-5
PSO		46.358	14.289	348.325
GWO		0.586	0.045	0.156
JS		0.295	0.026	0.045
2DME-SSA	$F_5(x)$	0	0	0
PSO		39.284	2.275	772.356
GWO		5.139	2.246	20.145
JS		4.953	0	2.094

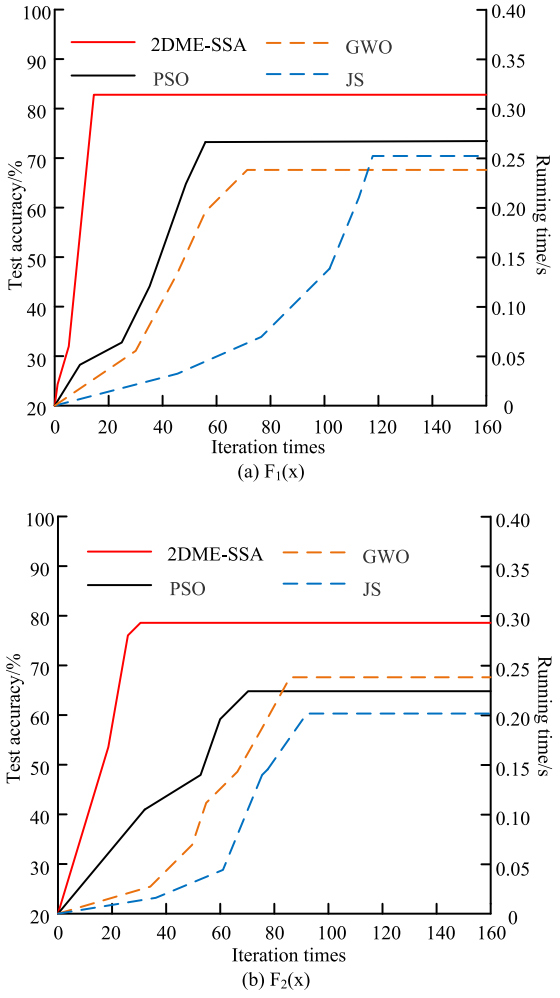


FIGURE 8. Comparison of running time and test accuracy of different algorithms.

In Figure 9 (a), 2DME-SSA achieved the optimal threshold for all four different images, especially in flower images with more vivid color contrast. The optimal threshold of 2DME-SSA was 22, 17, and 25 higher than the other three algorithms, respectively. In Figure 9 (b), in portrait segmentation, the optimal FV of JS algorithm was the same as that of 2DME-SSA. 2DME-SSA achieved the optimal FV in other images, indicating that 2DME-SSA has excellent convergence and robustness. The PSNR comparison of different algorithms is shown in Figure 10.

In Figure 10 (a), during normal IS, the PSNR of 2DME-SSA approached convergence at 11.4 after 80 iterations, while the other three algorithms approached convergence at 94, 116, and 121 iterations, respectively. The maximum PSNR of 2DME-SSA was 0.5, 1.1, and 1.3 higher than other algorithms, respectively. In Figure 10 (b), during medical IS, the PSNR of 2DME-SSA approached convergence after 84 iterations, and the convergence time was not significantly different from other algorithms. The maximum PSNR was 1.4, 1.5, and 1.7 higher than other algorithms, respectively. The comparison of feature similarity of images segmented by different algorithms is shown in Figure 11.

In Figure 11 (a), for flower images with distinct colors, the 2DME-SSA segmented image had the highest feature similarity, reaching 68.5%, which was 2.3%, 5.4%, and 7.2% higher than the other three algorithms, respectively, and the convergence speed of 2DME-SSA was faster. In Figure 11 (b), for more complex portraits, the maximum similarity of image features after 2DME-SSA segmentation was 63.8%, which was 3.4%, 2.7%, and 5.2% higher than other algorithms, respectively. 2DME-SSA tended to converge after 43 iterations.

In Figure 12 (a), for medical images with greater noise interference, the maximum feature similarity of the pulmonary nodule images segmented by 2DME-SSA was 74.1%, which was 5.7%, 7.5%, and 10.8% higher than other algorithms, respectively. The convergence speed was about 37 iterations slower than that of ordinary images. In Figure 12 (b), the maximum feature similarity of the

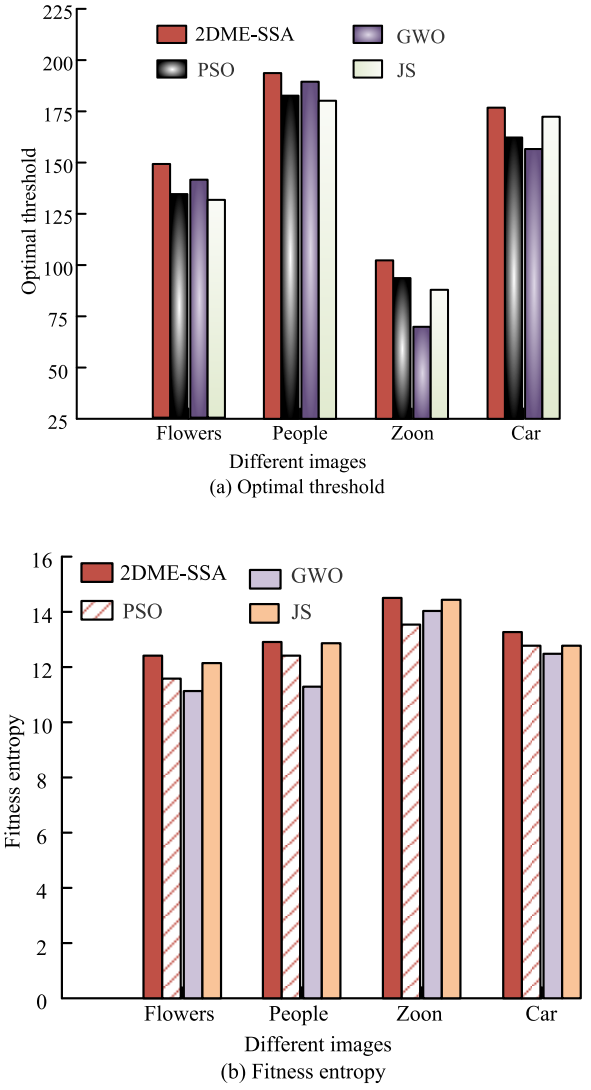


FIGURE 9. Optimal thresholds and fitness values of different algorithms.

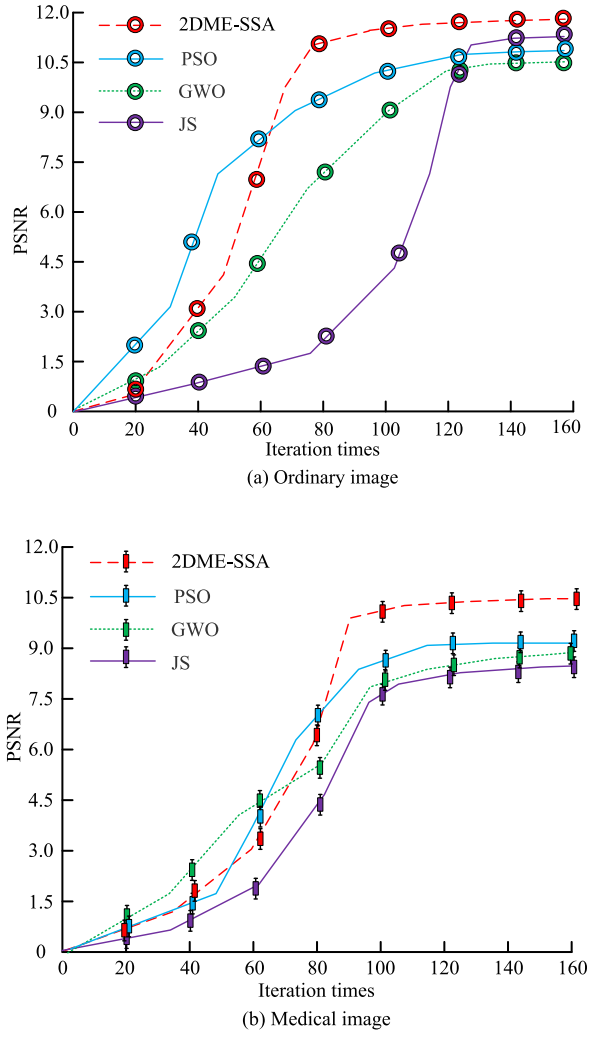


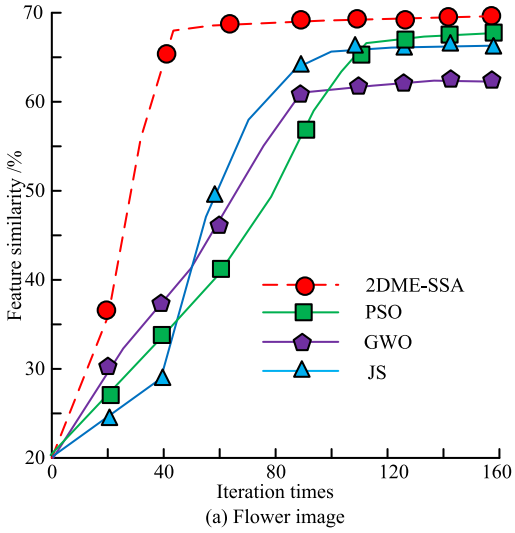
FIGURE 10. PSNR of different algorithms.

pulmonary shadow image segmented by 2DME-SSA was 73.4%, which was 7.9%, 10.2%, and 13.7% higher than other algorithms, respectively. It tended to converge after 76 iterations. The comparison of the segmentation results of the three algorithms is shown in Figure 13.

In Figure 13, a, b, and c represent the GT mask comparison of lung images, and d, e, and f represent the segmentation mask comparison. In different types of IS experiments, the 2DME-SSA segmented images had the maximum feature similarity, clearer boundaries, and the smallest difference from the real situation. The 2DME-SSA algorithm had the smallest difference between predicted and real masks, and the highest pixel accuracy. The lung lesion area of 2 DME-SSA algorithm was more complete, the image lesions segmented by PSO algorithm were missing, and the lesion loss of JS algorithm was more obvious. The study manually labeled the lung images and compared the segmentation results with those of 2DME-SSA, and the algorithm achieved a segmentation accuracy of 78.6%. The segmentation accuracy is 16.4% and 21.5% higher than that of PSO and JS algorithms, respectively.

To analyze the computational complexity of SSA-SOM algorithm and SOM algorithm, the time required by different algorithms to process the same amount of data under the same hardware conditions was measured experimentally. The results of the time complexity comparison of the three algorithms are shown in Table 4.

In Table 4, the average processing times of the 2DME-SSA algorithm were 34.2 ms, 78.5 ms, and 177.8 ms when the sample sizes were 1000, 2000, and 3000, respectively, which were lower than those of the PSO algorithm corresponding to 39.6 ms, 120.3 ms, and 268.4 ms, and those of the GWO algorithm corresponding to 41.5 ms, 161.4 ms, and 374.8 ms. The results showed that the 2DME-SSA algorithm had lower computational complexity and higher efficiency than the PSO algorithm and the GWO algorithm



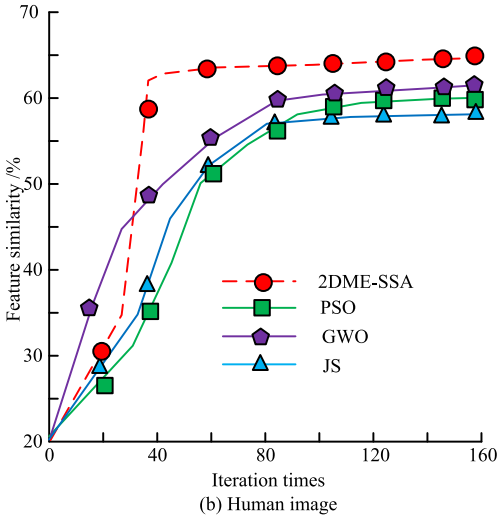
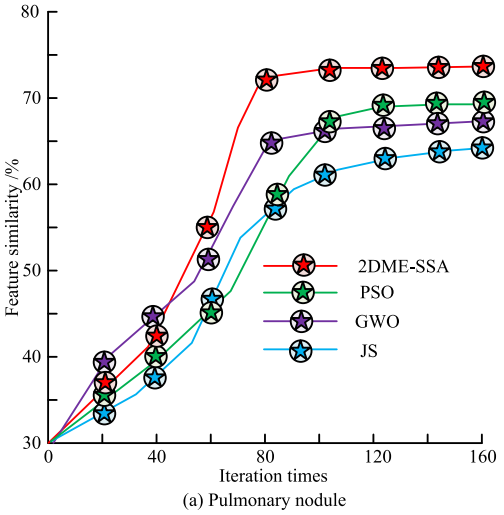


FIGURE 11. Feature similarity of common images segmented by different algorithms.

TABLE 4. The time complexity comparison of the three algorithms.

Sample size	Model type	Average processing time (ms)	Standard deviation (ms)	Time complexity $O(n^2)$ evaluation
1000	2DME-SSA	34.2	2.2	Lower
	PSO	39.6	2.8	Normal
	GWO	41.5	3.5	Normal
	SSA-SOM	78.5	5.2	Lower
2000	PSO	120.3	7.7	Normal
	GWO	161.4	8.2	Higher
	SSA-SOM	177.8	8.3	Lower
3000	PSO	268.4	11.9	Normal
	GWO	374.8	12.5	Higher

when processing the same amount of data. Meanwhile, with the increase of the number of samples, the time complexity of all three algorithms showed an increasing trend of $O(n^2)$, but the growth rate of 2DME-SSA algorithm was relatively low.



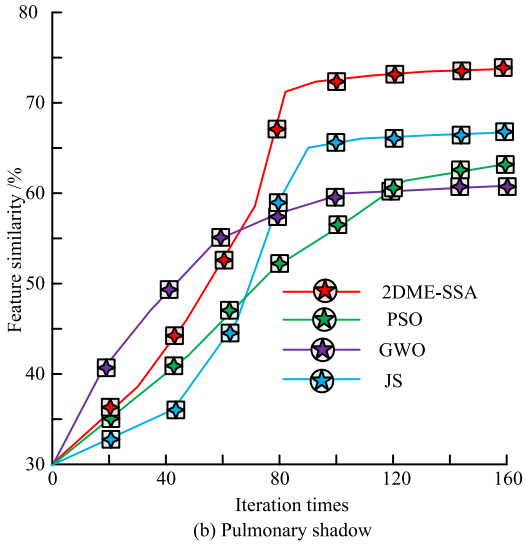


FIGURE 12. Feature similarity of pulmonary images segmented by different algorithms.

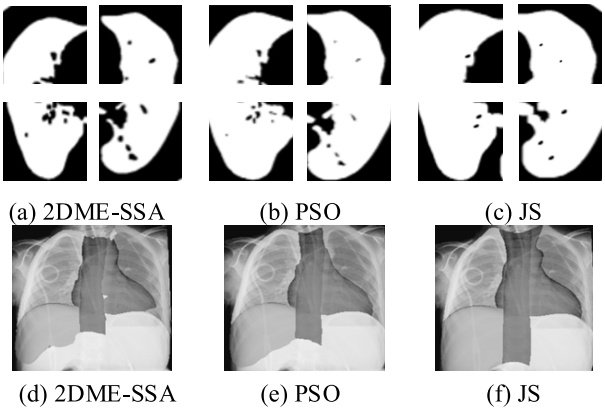


FIGURE 13. Comparison of segmentation results of three algorithms.

IV. DISCUSSION AND CONCLUSION

Targeting at the issues of low segmentation accuracy and poor feature similarity in traditional IS methods, an improved

GA and optimized 2DME IS method were proposed. The research results showed that the improved GA for medical image optimization curve had a faster descent speed, with fitness decreasing from 870 to 41 in the first 20 iterations, an average of 77 times faster than other algorithms. The feature similarity of brain images was on average 13.3% higher than other algorithms, and the feature similarity of heart images was on average 10.3% higher than other algorithms. The segmentation accuracy of the improved GA was less affected by noise, with an average decrease of 2.2% in segmentation accuracy after noise interference. The optimized 2DME IS method could obtain the global OS in different test functions, and the optimized algorithm had a faster convergence curve, averaging about 80 times faster than other algorithms. The optimized method had an average threshold and FV that were 21.3 and 3.7 higher than other algorithms, respectively. The PSNR convergence speed of 2DME-SSA was on average 110.3 times faster than other algorithms, and the maximum PSNR was on average 0.96 higher than other algorithms. The feature similarity of 2DME-SSA segmentation of medical images was the highest, averaging 8% higher than other algorithms, and the convergence speed was also faster. The improved algorithm can effectively enhance the feature similarity and PSNR of segmented images, with faster convergence speed and significantly improved resistance to noise interference. There are still some shortcomings in this study, such as the algorithm running for a long time and the need to shorten the time spent on each iteration in the future.

REFERENCES

[1] C. Shen and K. Zhang, "Two-stage improved grey wolf optimization algorithm for feature selection on high-dimensional classification," *Complex Intell. Syst.*, vol. 8, no. 1, pp. 2769–2789, Aug. 2022, doi: 10.3390/s20154065.

[2] S. Kak, "Self-similarity and the maximum entropy principle in the genetic code," *Theory Biosciences*, vol. 142, no. 3, pp. 205–210, Sep. 2023, doi: 10.1007/s12064-023-00396-y.

[3] A. Wunnava, M. Kumar Naik, R. Panda, B. Jena, and A. Abraham, "A differential evolutionary adaptive Harris hawks optimization for two dimensional practical Masi entropy-based multilevel image thresholding," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 34, no. 6, pp. 3011–3024, Jun. 2022, doi: 10.1016/j.jksuci.2020.05.001.

[4] A. Sohail, "Genetic algorithms in the fields of artificial intelligence and data sciences," *Ann. Data Sci.*, vol. 10, no. 4, pp. 1007–1018, Aug. 2023, doi: 10.1007/s40745-021-00354-9.

[5] M. Ince, "Automatic and intelligent content visualization system based on deep learning and genetic algorithm," *Neural Comput. Appl.*, vol. 34, no. 3, pp. 2473–2493, Feb. 2022, doi: 10.1007/s00521-022-06887-1.

[6] A. G. Hussien, A. A. Heidari, X. Ye, G. Liang, H. Chen, and Z. Pan, "Boosting whale optimization with evolution strategy and Gaussian random walks: An image segmentation method," *Eng. Comput.*, vol. 39, no. 3, pp. 1935–1979, Jun. 2023, doi: 10.1007/s00366-021-01542-0.

[7] G. Cheng, C. Lang, and J. Han, "Holistic prototype activation for few-shot segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4650–4666, Apr. 2023, doi: 10.1109/TPAMI.2022.3193587.

[8] C. Lang, G. Cheng, B. Tu, C. Li, and J. Han, "Base and meta: A new perspective on few-shot segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10669–10686, Apr. 2023, doi: 10.1109/TPAMI.2023.3289049.

[9] C. Lang, G. Cheng, B. Tu, and J. Han, "Few-shot segmentation via Divide-and-Conquer proxies," *Int. J. Comput. Vis.*, vol. 132, no. 1, pp. 261–283, Jan. 2024, doi: 10.1007/s11263-023-01886-8.

[10] M. Abdel-Basset, R. Mohamed, and M. Abouhawwash, "Hybrid marine predators algorithm for image segmentation: Analysis and validations," *Int. J. Environ. Res. Public Health.*, vol. 55, no. 4, pp. 3315–3367, Apr. 2022, doi: 10.1007/s10462-021-10086-0.

[11] F. S. Gharehchopogh and T. Ibrikci, "An improved African vultures optimization algorithm using different fitness functions for multi-level thresholding image segmentation," *Multimed Tools Appl.*, vol. 83, no. 6, pp. 16929–16975, Feb. 2024, doi: 10.1007/s11042-023-16300-1.

[12] A. A. Khan, A. A. Shaikh, Z. A. Shaikh, A. A. Laghari, and S. Karim, "IPM-model: AI and metaheuristic-enabled face recognition using image partial matching for multimedia forensics investigation with genetic algorithm," *Multimedia Tools Appl.*, vol. 81, no. 17, pp. 23533–23549, Jul. 2022, doi: 10.1007/s11042-022-12398-x.

[13] D. Wu and C. Yuan, "Threshold image segmentation based on improved sparrow search algorithm," *Multimedia Tools Appl.*, vol. 81, no. 23, pp. 33513–33546, Sep. 2022, doi: 10.1007/s11042-022-13073-x.

[14] B. Alhijawi and A. Awajan, "Genetic algorithms: Theory, genetic operators, solutions, and applications," *Evol. Intell.*, vol. 17, no. 3, pp. 1245–1256, Jun. 2024, doi: 10.1007/s12065-023-00822-6.

[15] A. A. Rafique, M. Gochoo, A. Jalal, and K. Kim, "Maximum entropy scaled super pixels segmentation for multi-object detection and scene recognition via deep belief network," *Multimedia Tools Appl.*, vol. 82, no. 9, pp. 13401–13430, Apr. 2023, doi: 10.1007/s11042-022-13717-y.

[16] L. Zhai and S. Feng, "A novel evacuation path planning method based on improved genetic algorithm," *J. Intell. Fuzzy Syst.*, vol. 42, no. 3, pp. 1813–1823, Feb. 2022, doi: 10.3233/jifs-211214.

[17] J. Purohit and R. Dave, "Leveraging deep learning techniques to obtain efficacious segmentation results," *Arch. Adv. Eng. Sci.*, vol. 1, no. 1, pp. 11–26, Jul. 2023, doi: 10.47852/bonviewaaes32021220.

[18] I. Kumar, C. Bhatt, and K. U. Singh, "Entropy based automatic unsupervised brain intracranial hemorrhage segmentation using CT images," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 34, no. 6, pp. 2589–2600, Jun. 2022, doi: 10.1016/j.jksuci.2020.01.003.

[19] A. Sharma, R. Chaturvedi, and A. Bhargava, "A novel opposition based improved firefly algorithm for multilevel image segmentation," *Multimedia Tools Appl.*, vol. 81, no. 11, pp. 15521–15544, May 2022, doi: 10.1007/s11042-022-12303-6.

[20] Z. Z. Wang and S. H. Goh, "A maximum entropy method using fractional moments and deep learning for geotechnical reliability analysis," *Acta Geotechnica*, vol. 17, no. 4, pp. 1147–1166, Apr. 2022, doi: 10.1007/s11440-021-01326-2.

[21] Y. Han, W. Chen, A. A. Heidari, and H. Chen, "Multi-verse optimizer with Rosenbrock and diffusion mechanisms for multilevel threshold image segmentation from COVID-19 chest X-ray images," *J. Bionic Eng.*, vol. 20, no. 3, pp. 1198–1262, May 2023, doi: 10.1007/s42235-022-00295-w.

[22] B. Zhang and H. Wang, "Exploring the advantages of the maximum entropy model in calibrating cellular automata for urban growth simulation: A comparative study of four methods," *GIScience Remote Sens.*, vol. 59, no. 1, pp. 71–95, Dec. 2022, doi: 10.1080/15481603.2021.2016240.

[23] N. Dhanachandra, Y. J. Chanu, and K. M. Singh, "A new hybrid image segmentation approach using clustering and black hole algorithm," *Comput. Intell.*, vol. 39, no. 2, pp. 194–213, Apr. 2023, doi: 10.1111/coin.12297.

[24] S. S. Ittannavar and R. H. Havaldar, "Detection of breast cancer using the infinite feature selection with genetic algorithm and deep neural network," *Distrib. Parallel Databases*, vol. 40, no. 4, pp. 675–697, Dec. 2022, doi: 10.1007/s10619-021-07355-w.

[25] A. Yang, Y. Bai, and L. Kangkang, "Application of SVM and its improved model in image segmentation," *Distrib Parallel Databases.*, vol. 27, no. 3, pp. 851–861, Jun. 2022, doi: 10.1007/s11036-021-01817-2.

[26] H. Yazid, S. N. Basah, S. A. Rahim, M. J. A. Safar, and K. S. Basaruddin, "Performance analysis of entropy thresholding for successful image segmentation," *Multimedia Tools Appl.*, vol. 81, no. 5, pp. 6433–6450, Feb. 2022, doi: 10.1007/s11042-021-11813-z.

[27] R. Rai, A. Das, and K. G. Dhal, "Nature-inspired optimization algorithms and their significance in multi-thresholding image segmentation: An inclusive review," *Evolving Syst.*, vol. 13, no. 6, pp. 889–945, Dec. 2022, doi: 10.1007/s12530-022-09425-5.

[28] K. M. Hosny, A. M. Khalid, H. M. Hamza, and S. Mirjalili, "Multilevel thresholding satellite image segmentation using chaotic coronavirus optimization algorithm with hybrid fitness function," *Neural Comput. Appl.*, vol. 35, no. 1, pp. 855–886, Jan. 2023, doi: 10.1007/s00521-022-07718-z.

[29] C. Dong, T. Jian, and F. Weisong, "Customer segmentation using K-means clustering and the hybrid particle swarm optimization algorithm," *Comput. J.*, vol. 66, no. 4, pp. 941–962, Apr. 2023, doi: 10.1016/j.asoc.2021.107924.
[30] M. Abdel-Basset, R. Mohamed, and M. Abouhawwash, "A new fusion of whale optimizer algorithm with Kapur's entropy for multi-threshold image segmentation: Analysis and validations," *Artif. Intell. Rev.*, vol. 55, no. 8, pp. 6389–6459, Dec. 2022, doi: 10.1007/s10462-022-10157-w.



JIN WANG was born in Heshun, Shanxi, China, in 1987. She received the bachelor's degree in electronic information and science technology from Shanxi Normal University, in 2007, and the master's degree in communication and information systems from Yanshan University, in 2010. She is currently pursuing the Ph.D. degree with the College of Computing, Informatics and Mathematics, Universiti Teknologi MARA (UiTM), Shah Alam, Malaysia. She is an Associate Professor with Taiyuan Institute of Technology, China. She has published several articles that have been included by SCI and EI databases. Her research interest includes medical image segmentation.



YANLI TAN was born in Taiyuan, Shanxi, China, in 1978. She received the bachelor's degree in applied electronic technology from North China Institute of Technology, in 2002, and the master's degree in signal and information processing from the North University of China, in 2008. She is currently pursuing the Ph.D. degree with the College of Computing, Informatics and Mathematics, Universiti Teknologi MARA (UiTM), Shah Alam, Malaysia. She has published several articles that have been included by SCI and EI databases. Her research interests include deep learning and anomaly detection.



XIAONING BO was born in Xinzhou, Shanxi, China, in 1987. He received the bachelor's degree in electronic information engineering from Taiyuan University of Technology, in 2009, and the master' degree in communication and information systems from Beijing Jiaotong University, in 2014. He is currently pursuing the Ph.D. degree with the North University of China. He is a Lecturer with Taiyuan Institute of Technology. He has published several articles that have been included by SCI and EI databases. His research interests include computational imaging and image processing.



GUOQIN LI was born in Shanxi, in 1992. She received the bachelor's degree from North University of China and the master's degree in signal and information processing from Northwestern Polytechnical University. She is currently pursuing the Ph.D. degree with the College of Computing, Informatics and Mathematics, Universiti Teknologi MARA (UiTM), Shah Alam, Malaysia. She is a Lecturer with Taiyuan Institute of Technology. She has published several articles that have been included by SCI and EI databases. Her current research interest includes medical image processing based on deep learning.